

Curriculum Vitae

Pratush Brahma, PhD

University of Florida
Dept. of Zoology
PO Box 110430
Gainesville, FL 32611-1234 USA

pratush.brahma@ufl.edu
+1 (352) 745-9009
Twitter: @BrahmaPratush
Github: pratushbrahma

Education

2018-2026	Ph.D. in Biological Sciences, University of Florida
2013-2018	Integrated MSc in Biology, National Institute of Science Education and Research, Bhubaneswar

Professional Positions

Summer 2026	Program Director , Center for Precollegiate Education and Training, University of Florida Supervisor: Dr. Hazel Levy
2022-2026	Doctoral Researcher , Entomology Department, University of Florida Supervisor: Dr. Lisa Taylor Project: "Investigating the independent evolution of color vision across different species of jumping spiders"
2018-2026	Doctoral Researcher , Department of Biology, University of Florida Supervisor: Dr. St. Mary Colette Project: "Predicting density-dependent variation in reproductive strategies through Dynamic State Variable Modeling"
2016-2018	Masters Research Student , School of Biological Sciences, NISER, India Supervisor: Dr. Manjusha Dixit Project: "Understanding role of FRG1 in cancer angiogenesis using in-vivo and ex-ovo systems"
Summer 2016	Summer Intern , Center for Ecological Sciences, IISc, India Supervisor: Dr. Kavita Isvaran Project: "Understanding intrasexual competition in rock agamas by studying their mating strategies and spatial patterns"
Summer 2015	Summer Intern , School of Biological Sciences, NISER, India Supervisor: Dr. Manjusha Dixit Project: "Genetic association studies of gall bladder carcinoma in Eastern Indian Population"
Summer 2014	Summer Intern , Center for Ecological Sciences, IISc, India Supervisor: Dr. Karthik Shanker Project: "Biodiversity Assessment of Anurans (Microhylidae) in Western Ghats"

Publications

1. Naik, M.⁺, Brahma, P.⁺, & Dixit, M. (2018). A cost-effective and efficient chick ex-ovo CAM assay protocol to assess angiogenesis. *Methods and protocols*, 1(2), 19.
⁺These authors contributed equally to the research and the publication
2. Mukherjee, B., Brahma, P., Mohapatra, T., Chawla, S., & Dixit, M. (2023). Reduced FRG1 expression promotes angiogenesis via activation of the FGF2-mediated ERK/AKT pathway. *FEBS Open bio*, 13(5), 804-817.

Manuscripts in Prep

1. Brahma, P., Lietzenmayer, L., Menon, Y., Colette, SM., Uma, D., & Taylor, L. (2024). The use of color in foraging in Lessert's Rainbow Spider (*Stenaelurillus lesserti*)

2. Brahma, P., & St. Mary, CM. (2024). Predicting state-dependent variation in male reproductive strategies based on changes in population density
3. Brahma, P., & St. Mary, CM. (2024). Predicting the influence of population density on female reproductive tactics using an optimization approach
4. Brahma, P., & St. Mary, CM. (2024). Population viability analysis of density-dependent variation in reproduction behavior
5. Lietzenmayer, L., Persad, A., Stone, K., Krausman, Z., Willer, A., Faustino, C., Brahma, P. & Taylor, L. (2024). Does color vision improve jumping spiders' ability to find colored prey on colored backgrounds?

Conferences & Workshops

- September 2025 **Presenter**, Southeastern Conference for Animal Behavior, University of Tennessee, Knoxville, USA
Poster Title: *Reproductive Responses to Changing Densities: A Potential Mechanism for Successful Invasion*
- July 2025 **Presenter**, Animal Behavior Symposium, Baltimore, Animal Behavior Society
Talk Title: *Understanding Density-Dependent Reproductive Behavior and Its Impact on Population Dynamics for Conservation Management*
Session: *Behavior and Conservation*
- Fall 2024 Preparing Future Faculty Program **Workshop Series**, Center for Teaching Excellence, University of Florida
- May 2024 Supporting Inclusion in Graduate Networks and Leadership (SIGNAL) **Workshop**, Virtual, Animal Behavior Society
- March 2024 Mentorship - Matching **Workshop**, Virtual, Animal Behavior Collective
- October 2023 **Presenter**, Student Conference on Conservation Science, American Museum of Natural History, New York, USA
Poster Title: *Density- Dependent Variation in Reproductive Behavior Across Invasion Phases: A Dynamic State Variable Approach to Invasive Species Management*
- August 2021 **Presenter**, Animal Behavior Symposium, Virtual, Animal Behavior Society
Talk Title: *Predicting Density-Dependent Patterns of Reproduction using Dynamic State Variable Models*
Session: *Sexy Time: Mating Behavior and Sexual Selection*
- October 2017 **Participant**, NextGen Genomics, Biology, Bioinformatics and Technologies (NGBT) Conference, SciGenom Research Foundation, Bhubaneswar, India
- September 2017 **Participant**, Genomics Analysis and Technology Conference, National Institute for Science Education and Research, India
- March 2016 **Participant**, PAI Conference and International Symposium on Stress, Microbiomes and Probiotics, National Institute for Science Education and Research, India

Research Skills

Laboratory	Behavioral Assays, Experimental Design, Microscopy, PCR, Immunohistochemistry, Cloning, Western Blotting, Cell Culture, RFLP, Mice Matrigel and Tumor Assays, Chick-CAM Angiogenesis Assays, DNA Sequencing, Cryo-Sectioning
Field	Animal Handling: Mice, Lizards, Fish & Spiders, Animal Life History and Vital Measurements, Photography, Spatial Mapping, Trekking
Computational	Programs: R, Shell, Markup: Rmd, LATEX, Modeling: Dynamic State Variable Models, Agent/Individual-based Models, Population Dynamics Models
Analytical	Jump, QGIS, MEGA, ImageJ, PyMol, Snapgene

Grants and Achievements

2026	Animal Behavior Society Diversity Grant
2025	Southeastern Conference for Animal Behavior Travel Grant
2025	Animal Behavior Society Travel Grant
2025	Graduate Student Council Travel Grant, University of Florida
2023	College of Liberal Arts and Sciences Travel Grant, University of Florida
2018	Secured All India Rank 55 in Graduate Aptitude Test (GATE- Biology) and Qualified for the Doctoral Fellowship in India
2018	Secured All India Rank 68 in National Entrance Test (NET- Biology) and Qualified for the Doctoral Fellowship in India
2013-2018	Innovation in Science Pursuit for Inspire Research (INSPIRE) Scholarship by Ministry of Human Resource Development (MHRD) for the entire duration of Integrated Masters Degree
2013	Secured All India Rank 181 in National Entrance Screening Test, India

Teaching Experience

Teaching Assistant	Undergraduate Courses at University of Florida: Evolution (Fall 2023, Summer 2023, Summer 2022, Summer 2020, Fall 2019) Genetics (Fall 2022, Fall 2021, Spring 2021, Fall 2020, Spring 2020) Integral Principal Biology 2 Lab (Summer 2024, Spring 2024, Spring 2023, Summer 2021, Summer 2019, Spring 2021, Fall 2018) Integral Principal Biology 1 Lab (Fall 2024, Spring 2022)
Course Coordinator	High School Courses at University of Florida: Precollegiate Scholars (Summer 2024, Summer 2025)
Science Instructor	Middle School Courses at University of Florida: JumpStart (Summer 2024), Gator Labs (Summer 2024)
Counselor	Primary School Subjects at rural areas in Bhubaneswar, India

Outreach Activities

February 2026	Competition Judge , for the Florida STEM Symposium at University of Florida, Florida
September 2023	Spider Awareness , at Not-so-scary science at the Florida Museum of Natural History, Florida
February 2024	Guest Discussion , creating a Public Health awareness lecture on Ithink Biology's NPTEL Online Course at Azim Premji University, India

References

Dr. CM St. Mary

Program Director, Behavioral Systems Cluster
Integrative and Organismal Systems, BIO Directorate
National Science Foundation
cstmary@nsf.gov

Dr. Hazel Levy

Associate Director, Center for Precollegiate Education & Training
University of Florida
hclevy@ufl.edu

Dr. Lisa Taylor

Assistant Research Scientist, Department of Entomology
University of Florida
lisa.taylor@ufl.edu

Dr. Divya Uma

Professor, Department of Biology
Azim Premji University, Bengaluru, India
divya.uma@apu.edu.in

Dr. Rebecca Kimball

Professor, Department of Biology
University of Florida
rkimball@ufl.edu